RATING METHODOLOGY

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Rating Methodology

Construction

This methodology is no longer in effect. For more information on rating methodologies used by Moody's Investor Services, visit https://ratings.moodys.com/rating-methodologies.

This rating methodology replaces the *Construction* methodology published on 10 September 2021. While this methodology reflects many of the same core principles as the 2021 methodology, we have made changes to the scorecard to provide greater consistency of approach across our corporate sector methodologies. In the Scale factor, we have removed EBITA as a sub-factor. In the Business Profile factor, we have renamed the Expected Revenue & Margin Stability sub-factor as Revenue and Earnings Stability and moved it to a new Profitability and Efficiency factor. In the Leverage and Coverage factor, we have replaced the sub-factors EBITA/Interest Expense and FFO/Debt with EBITDA/Interest Expense and RCF/Net Debt, respectively. We have changed some factor and sub-factor weights, sub-factor descriptions, and scoring thresholds. We have also made limited editorial updates to some sections of the methodology, and we have changed the presentation of the scorecard.

OUTDATED METHODOLOGY

Scope

This methodology applies to companies globally that are primarily engaged in the construction or refurbishment of civil infrastructure (tunnels, roads, bridges, harbors); industrial infrastructure (manufacturing plants, power generating plants, oil and gas processing facilities); and buildings for commercial purposes (offices, warehouses) or public purposes (schools, hospitals, government buildings). These companies include general contractors or subcontractors.

This methodology also applies to companies that provide installation, repair and maintenance of electrical and mechanical systems, and power generation and telecommunications infrastructure.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

This methodology does not apply to building materials companies, companies focused on homebuilding for private customers, real estate development companies, and companies focused on the production and installation of heavy equipment and machinery for industrial users and utilities, which are rated using our other methodologies.[1]

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Sector overview

Cyclicality

The pattern of general economic cycles is a significant driver of demand for construction and refurbishment activity. Exposure to certain geographical areas, customers and end-markets determines the trajectory of a company's operating performance in relation to the cycles of specific economies. From a broader customer segmentation perspective, private customers and public customers exhibit different contracting behavior through the cycle. Private sector customers, whose contracts are normally of higher profitability, may delay or cancel new projects when the economy slows, but could shift towards demand for refurbishment projects depending on their specific financial condition and investment policies. Public sector contracts, which are typically less profitable, are closely tied to a government's fiscal policy or spending plans, which are not necessarily linked to the stage of an economic cycle in a specific country. Public sector spending could even show anti-cyclical spending patterns, supported by efforts to stimulate the economy through the construction sector.

Relatively low margins and volatile earnings

Low industry profitability levels are often a result of extensive levels of subcontracting, which reduce the value added revenues and hence profitability for a company but substantially increase the level of operating flexibility through the cycle. In order to offer a complete solution package, many construction companies act as "general contractors" and coordinate resources from various "subcontractors" to execute the work on a project, while retaining the responsibility for project completion. Subcontracting allows them to have a more flexible cost structure, but leads to substantially reduced profitability. Since the general contractor bills the whole project, the general contractor will recognize the full revenues but pass on part of the profit to the subcontractors.

Another determining factor of profitability and earnings volatility is the type of construction contracts. Fixed-price contracts typically offer high margin potential, but the risk of cost overruns on complex projects or those with a long project duration is more severe than for cost-plus contracts or fee for service contracts. These contracts do not expose contractors to the risk of cost overruns. Therefore, these cost-plus type contracts offer a higher level of earnings stability but provide lower margin potential.

Profitability levels in the industry are also burdened by time-consuming competitive tender procedures for order awards, leading to aggressive bidding by construction companies. In highly competitive markets or in cyclical downturns, companies may relax margin standards in order to win tenders and fill capacity with the knowledge that variable costs (but not all fixed costs) will be covered. This behavior can lead to operating losses and cash flow deficits. Public contracts in particular are normally secured through these tender procedures and are generally linked to greater risk and reduced margins given the focus on price compared to more profitable referral work from private customers.

Effective risk management is essential to project profitability

One of the key drivers of a project's profitability is the bidding process. The competitive nature of the bidding process can drive companies to be aggressive and expose them to the risks of adverse surprises.

Project construction risks include (i) cost overruns above an initial construction budget, which are a greater threat in the case of fixed-price contracts; and (ii) time overruns in the case of delayed completion, which could be a result of an unreasonable construction schedule or a project's complexity. Project construction risks depend on the experience/track record of the contractor with the particular type of project being undertaken, and the degree to which local economic conditions around the project site will affect the cost and the availability of labor and materials over the construction period. Large projects could be exposed to regulatory or political risks in countries whose political, economic and legal systems are less predictable or are potentially unfavorable for companies in this sector.

Substantial working capital needs require financial flexibility

It is often necessary for construction companies to maintain financial flexibility. The construction industry sees large working capital swings not only due to payments received upon completion of major milestones for long-term projects, but also due to the utilization of a complex array of advance payments made by the customer. Certain customers and/or projects typically include payments in advance of services rendered or for the procurement of equipment and materials. A normal trend for these projects is to have higher cash balances during the initial phases of execution which then level out toward the end of the construction phase. As a result, a company's cash position is reduced as customer advances are worked off, unless they are replaced by advances on other projects. However, some customers and project types do not involve advance payments and require significant working capital investments

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early in the project life cycle. In addition, contractors can be subject to major demands on cash for working capital because of delayed payments from customers or due to sudden, large changes in project schedules. It is sometimes difficult to assess the underlying cash flows of construction companies since their operating results may not be an accurate indicator of their cash flows due to the use of percentage of completion accounting. When changes in project estimates occur, then the estimated impact on the entire project is recognized in the period in which they are determined. For example, if a project estimate is revised and the project is expected to result in a loss for the company, then the full amount of the loss is recognized in the period when they are discovered. Therefore, the percentage of completion method will often result in periods when reported revenues and operating earnings are poor predictors of the company's cash flows.

Rating approach

In this rating methodology, we explain our general approach to assessing credit risk of issuers in the construction industry globally, including the qualitative and quantitative factors that are likely to affect rating outcomes in this sector. We seek to incorporate all material credit considerations in ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

The following schematic illustrates our general framework for the analysis of construction companies, which includes the use of a scorecard. The scorecard-indicated outcome is not expected to match the actual rating for each company. For more information, see the "Other considerations" and "Limitations" sections.

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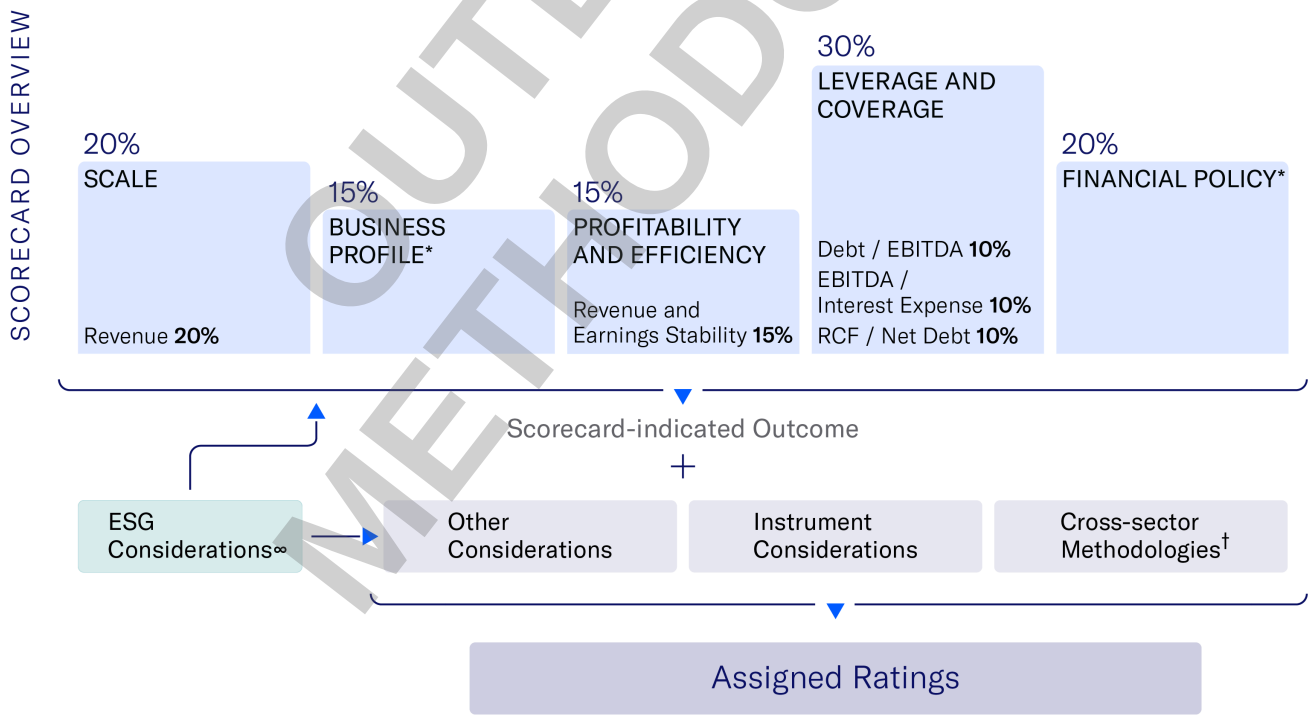
Exhibit 1
Illustration of the construction methodology framework



* This factor has no sub-factors.
∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.
† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.
Source: Moody's Ratings

Construction scorecard

For general information about how we use the scorecard and for a discussion of scorecard mechanics, please see the "Using the scorecard to arrive at a scorecard-indicated outcome" section. The scorecard does not include or address every factor that a rating committee may consider in assigning ratings in this sector. Please see the "Other considerations" and "Limitations" sections.

Exhibit 2

Construction scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Scale (20%)									
Revenue (USD Billion)[1]	20%	≥ $60	$30 - $60	$15 - $30	$8 - $15	$3 - $8	$1 - $3	$0.25 - $1	< $0.25
Factor: Business Profile (15%)									
Business Profile	15%	Extremely well diversified with several profitable global platforms that are market leaders in multiple continents.	Well diversified with several profitable global platforms and strong market positions in multiple continents.	A number of diversified segments, varying in size and profitability with just under half of revenues within one continent.	More than one balanced and profitable segment with the majority of revenues within one continent.	Heavily reliant on one segment with the significant majority of revenues within one continent, but diversified across several economic regions.	Heavily reliant on one segment with the significant majority of revenues within one continent and diversified across a few economic regions.	Heavily reliant on one segment and the significant majority of revenues expected to be within one economic region.	One segment generates all revenues and all sales expected to be within one small economic region.
Factor: Profitability and Efficiency (15%)									
Revenue and Earnings Stability	15%	Superior revenue, margin and backlog sustainability and technical capabilities that create very strong barriers to entry. Superior track record of project execution within scheduled time frame and established budget.	Extremely high revenue, margin and backlog sustainability and technical capabilities that create very strong barriers to entry. Very strong track record of project execution within scheduled time frame and established budget.	Very high revenue, margin and backlog sustainability and technical capabilities that create strong barriers to entry. Strong track record of project execution within scheduled time frame and established budget.	High revenue, margin and backlog sustainability and technical capabilities that create strong barriers to entry. Solid track record of project execution within scheduled time frame and established budget.	Good revenue, margin and backlog sustainability and technical capabilities that create barriers to entry. Generally good track record of project execution within scheduled time frame and established budget.	Moderate revenue, margin and backlog sustainability and technical capabilities that create modest barriers to entry. Moderate track record of project execution.	Volatile revenues, margins and backlog. Modest technical capabilities and limited competitive differentiation. Weak track record of project execution.	Volatile and unpredictable revenues, margins and backlog with no competitive differentiation. Poor track record of project execution.

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	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Leverage and Coverage (30%)									
Debt / EBITDA[2]	10%	≤ 0.5x	0.5x - 1x	1x - 1.5x	1.5x - 2.5x	2.5x - 4x	4x - 6x	6x - 8.5x	> 8.5x
EBITDA / Interest Expense[3]	10%	≥ 30x	20x - 30x	10x - 20x	7x - 10x	4x - 7x	2x - 4x	1x - 2x	< 1x
RCF / Net Debt[4]	10%	≥ 70%	50% - 70%	40% - 50%	25% - 40%	15% - 25%	8% - 15%	1% - 8%	< 1%
Factor: Financial Policy (20%)									
Financial Policy	20%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] For the linear scoring scale, the Aaa endpoint value is $100 billion. A value of $100 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[2] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The Ca endpoint value is 15x. A value of 15x or worse equates to a numeric score of 20.5, as does a negative Debt/EBITDA value.
[3] For the linear scoring scale, the Aaa endpoint value is 40x. A value of 40x or better equates to a numeric score of 0.5. The Ca endpoint value is 0x. A value of 0x or worse equates to a numeric score of 20.5.
[4] For the linear scoring scale, when net debt is positive, the Aaa endpoint value is 100%. A value of 100% or better equates to a numeric score of 0.5. The Ca endpoint value is 0%. A value of 0% or worse equates to a numeric score of 20.5. When net debt is negative and RCF is positive, the numeric score is 0.5. When net debt is negative and RCF is negative or zero, the numeric score is 20.5.
Source: Moody's Ratings

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Discussion of the scorecard factors

In this section, we explain our general approach for scoring each scorecard factor or sub-factor, and we describe why they are meaningful as credit indicators.

Factor: Scale (20% weight)

Why it matters

Scale is an important indicator of the overall depth of a company's business and its success in attracting a variety of customers, as well as its resilience to shocks, such as sudden shifts in demand or rapid cost increases. Scale can also provide a broader platform for sustainable earnings and cash flow generation and typically enhances a construction company's operating and financial flexibility and its ability to bid, finance and profitably execute large, long-term and complex projects. Large construction companies can accommodate a broad range of construction needs since they typically maintain a sizeable network of subcontractors and obtain various sources of financing, including bonding lines, which are key competitive advantages in the industry. In addition, scale in the construction industry often has a bearing on other key considerations such as geographic and segment diversity.

How we assess it for the scorecard

Scoring for this factor is based on one sub-factor: Revenue.

Revenue

Scale is measured (or estimated in the case of forward-looking expectations) using total reported revenue in US dollars.

Factor: Business Profile (15% weight)

Why it matters

The business profile of a construction company is important because it greatly influences its ability to generate sustainable earnings and operating cash flows. Market position and diversification across several continents or economic regions and exposure to a number of uncorrelated segments can mitigate earnings volatility, which can be affected by cyclical swings, changing levels of competition and project performance.

Geographic diversity is important because it (i) reduces a company's vulnerability to adverse economic shocks or cyclicality that may impact certain geographies; (ii) mitigates the impact of regional regulatory, environmental or safety issues; and (iii) provides exposure to different demand trends that may persist for an intermediate period of time in various regions. Geographic diversity usually tempers volatility by balancing slower and higher growth markets, regional economic swings, and seasonal or weather-related fluctuations in cash flows.

Segment diversity mitigates the impact of demand fluctuations, price competition and technological trends that can occur in particular segments.

How we assess it for the scorecard

Scoring for this factor is based on our assessment of the construction company's operational and geographic diversity and its position in the market.

We assess geographic diversity primarily based on the extent to which a construction company's revenues are spread across multiple continents, countries within a continent or economic regions within a country.

The effectiveness of segment diversity usually considers the correlation of individual segments. The definition of business segments may differ among companies, according to each company's strategic focus or construction services groupings. While the number of business segments discussed in annual reports usually serves as a good indicator of segment diversity, we may adjust these measures based on the estimated levels of correlation across the reported segments.

Factor: Profitability and Efficiency (15% weight)

Why it matters

Profits matter because they are needed to generate sustainable cash flow and maintain a competitive position, which includes making investments in operations.

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Revenue sustainability provides an indication of how protected a company is from short-term economic disruptions and of its future earnings potential.

Earnings stability is an indicator of the presence of sustainable competitive advantages (in particular if combined with evidence of stable market share). If a company is able to sustain a high level of profitability over a long period of time and no major competitors have emerged, that business is probably protected by significant barriers to entry that will help sustain high profitability in the future.

How we assess it for the scorecard
Scoring for this factor is based on one sub-factor: Revenue and Earnings Stability.

Revenue and Earnings Stability

Scoring for this factor is qualitative and based on a company's technical capabilities, its ability to create barriers to entry, and its track record of successful project execution. We assess a company's existing order backlog and the extent to which future orders are protected by competitive barriers. We recognize that many companies, particularly those serving utilities, have significant future revenue streams because of master service agreements. These future revenue streams are difficult to quantify and are usually not reported in backlog. Estimates of these revenue streams are often considered if sufficient information exists on which to base such estimates.

Factor: Leverage and Coverage (30% weight)

Why it matters
Leverage and coverage measures provide important indications of a company's financial flexibility and long-term viability. Strength in these measures is an indicator of a greater ability to make new investments, weather the vagaries of the business cycle and respond to unexpected challenges, which often occur in the construction industry given the periodic performance issues that arise.

This factor comprises three quantitative sub-factors:

Debt / EBITDA

The ratio of total debt to earnings before interest, taxes, depreciation and amortization (Debt/EBITDA) is an indicator of debt serviceability and financial leverage. The ratio is commonly used in this sector as a proxy for comparative financial strength.

EBITDA / Interest Expense

The ratio of earnings before interest, taxes, depreciation and amortization to interest expense (EBITDA/Interest Expense) is an indicator of a company's ability to meet its interest obligations.

RCF / Net Debt

The ratio of retained cash flow to net debt (RCF/Net Debt) is an indicator of a company's cash generation (before working capital movements and capital expenditures, and after dividend payments) relative to its net debt.

How we assess it for the scorecard
Scoring for this factor is based on three sub-factors: Debt/EBITDA; EBITDA/Interest Expense; and RCF/Net Debt.[2]

Debt / EBITDA

The numerator is total debt, and the denominator is EBITDA.

EBITDA / Interest Expense

The numerator is EBITDA, and the denominator is interest expense.

RCF / Net Debt

The numerator is retained cash flow, and the denominator is net debt.

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Factor: Financial Policy (20% weight)

Why it matters

Financial policy encompasses management and board tolerance for financial risk and commitment to a strong credit profile. It is an important rating determinant, because it directly affects debt levels, credit quality, the future direction for the company and the risk of adverse changes in financing and capital structure.

Financial risk tolerance serves as a guidepost to investment and capital allocation. An expectation that management will be committed to sustaining an improved credit profile is often necessary to support an upgrade. For example, we may not upgrade the ratings of a company that has built flexibility within its rating category if we believe the company will use that flexibility to fund a strategic acquisition, cash distribution to shareholders, spinoff or other leveraging transaction. Conversely, a company's credit rating may be better able to withstand a moderate leveraging event if management places a high priority on returning credit metrics to pre-transaction levels and has consistently demonstrated the commitment to do so through prior actions. Liquidity management is an important aspect of overall risk management and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which is discussed in the "Other considerations" section.

Construction companies have historically used acquisitions to spur revenue growth, expand business lines, consolidate market positions, advance cost synergies or seek to access new technologies or capabilities. The impact of an acquisition on a rating will depend on the company's existing capital structure, the degree to which it is changed by the acquisition and its focus on returning its credit metrics to a level that is appropriate for its rating.

How we assess it for the scorecard

We assess the issuer's desired capital structure or targeted credit profile, its history of prior actions, including its track record of risk and liquidity management, and its adherence to its commitments. Attention is paid to management's operating performance and use of cash flow through different phases of economic and industry cycles. Also of interest is the way in which management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges or regulatory pressures. Considerations include a company's public commitments in this area, its track record of adhering to commitments and our views on the ability of the company to achieve its targets.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential negative impact of M&A or other types of balance-sheet-transforming events. Management's appetite for M&A activity is assessed, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. Frequency and materiality of acquisitions and previous financing choices are evaluated. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider negative repercussions caused by shareholders' willingness to sell the company.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. A track record of favoring shareholder returns at the expense of debtholders is likely to be viewed negatively in scoring this factor.

Other considerations

Ratings may reflect consideration of additional factors that are not in the scorecard, usually because the factor's credit importance varies widely among the issuers in the sector or because the factor may be important only under certain circumstances or for a subset of issuers. Such factors can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends, can also affect ratings.

The following are examples of additional considerations that may be reflected in our ratings and that may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and sub-factors in the scorecard.

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Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given issuer is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, please see our methodology that describes our general principles for assessing ESG risks.[3]

Liquidity

Liquidity is an important rating consideration for all construction companies, and extremely weak liquidity can heavily affect ratings in many cases. However, the relative strength or weakness of liquidity may not have a substantial impact in discriminating between two issuers with an otherwise similar credit profile. Liquidity can be particularly important where companies have large short-term demands on liquidity, for example in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely near-term liquidity requirements from the perspective of both sources and uses of cash. For more information on our approach, please see our liquidity cross-sector methodology.[4]

Regulatory considerations

Companies in the construction sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage particular issuers.

Our view of future regulations plays an important role in our expectations of future financial metrics, as well as our confidence level in the ability of an issuer to generate sufficient cash flows relative to its debt burden over the medium and longer term. Regulatory considerations also play a role in our assessment of an issuer's business profile. In some circumstances, regulatory considerations may also be a rating factor outside of the scorecard, for instance when regulatory change is swift.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies and in evaluating management performance relative to performance of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that are far in excess of their operating needs. This excess cash, together with the company's policy and motivations, can be an important credit consideration.

The level of cash required to run a business can vary materially based on the region(s) of operation and the specific sub-sectors in which the issuer operates. As a result, cash on the balance sheet is most often considered qualitatively, by assessing the company's track record and financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We analyze possible uses of cash, especially for companies that may be under shareholder pressure to return it. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some beneficial cushion against credit deterioration, and cash balances are often considered in our analysis of near-term liquidity sources and uses. For more information, please see the "Liquidity" section.

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While the scorecard in this methodology uses certain leverage and coverage ratios with total (or gross) debt, in cases where we are sufficiently confident that cash will remain on the balance sheet or will be used for creditor-friendly purposes, we may consider excess cash in a more quantitative manner. We do not typically adjust the balance-sheet debt for any specific amount because this implies greater precision than we think is achievable for the uncertain future uses of cash. We may consider creditor-friendly use of cash in our forward view of metrics, for instance, when we expect the cash to be used for debt repayment. We may also cite rating threshold levels for certain companies based on net debt ratios, particularly when these issuers have publicly stated financial targets based on net debt metrics. In cases where we believe that cash on the balance sheet does not confer meaningful credit support, we are more likely to cite gross debt ratios in our credit analysis and rating threshold levels.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this industry; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important to our forward view of metrics that are in the scorecard or other rating factors.

Non-wholly owned subsidiaries

Some companies in the construction sector choose to dilute their equity stake in certain material subsidiaries, for example through an initial public offering, which may in some cases negatively impact future financial flexibility. While improving cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative impact on cash flows available to the parent company that may not be fully reflected in consolidated financial statements. For example, in the case of an equity stake reduction in a subsidiary down to 75%, in the parent's financial statements, all revenue and EBITDA of the subsidiary would typically still be consolidated at the group level.

The parent's share of dividend flows from a non-wholly owned subsidiary is reduced, and minority stakes can increase structural subordination, since dividend flows to minority interest holders are made before the cash flows are available to service debt at the parent company. While less frequent, sale of a minority stake may be accompanied by policies protective of the subsidiary that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. However, in some cases we may find that an additional view of financial results, such as analyzing cash flows on a proportional consolidation basis, may be very useful to augment our analysis based on consolidated financial statements. When equity dilution or structural subordination arising from non-wholly owned subsidiaries is material and negative, the credit impact is captured in ratings but may not be fully reflected in scorecard-indicated outcomes.

For companies that hold material minority interest stakes, consolidated funds from operations typically includes the dividends received from the minority subsidiary, while none of its debt is consolidated. When such dividends are material to the company's cash flows, these cash flows may be subject to interruption if they are required for the minority subsidiary's debt service, capital expenditures or other cash needs. When minority interest dividends are material, we may also find that proportional consolidation or another additional view of financial results is useful to augment our analysis of consolidated financials. We would generally also consider structural subordination in these cases. Proportional consolidation brings a portion of the minority subsidiary's debt onto the balance sheet, but this debt is structurally senior to debt at the parent company, because it is closer to the assets and cash flows of the minority subsidiary. When these credit considerations are material, their impact is captured in ratings but may not be fully reflected in scorecard-indicated outcomes.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in an issuer's fundamental creditworthiness, which may cause actual ratings to be lower than the scorecard-indicated outcome. Event risks — which are varied and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized firm. Some other types of event risks include M&A, asset sales, spinoffs, shareholder distributions, litigation, pandemics, significant cybercrime events and geopolitical conflicts.

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Parental support

Ownership can provide ratings lift for a particular company in the construction sector if it is owned by a highly rated owner(s) and is viewed to be of strategic importance to those owners. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the issuer in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or has already done so in the past. Conversely, if the parent puts a high dividend burden on the issuer, which in turn reduces its flexibility, the ratings would reflect this risk.

Government-related issuers may receive ratings uplift due to expected government support. However, for certain issuers, government ownership can have a negative impact on the underlying Baseline Credit Assessment.[5] For example, price controls, onerous taxation and high distributions can have a negative effect on an issuer's underlying credit profile.

Cyclical sectors

Scorecard-indicated outcomes in cyclical sectors may be higher than the rating at the top of the economic cycle and lower than the rating at the bottom of the cycle. While using annual financials in the scorecard typically provides very useful insights into recent or near-term results, ratings may also reflect our expectations for the progression of yearly results over a longer period that may include a full economic cycle. However, cyclicality itself poses many different types of risks to companies, and cycles do not reverse themselves with predictable regularity. A cyclical sector may also be affected by a secular decline or expansion. These considerations may be incorporated qualitatively in ratings.

Using the scorecard to arrive at a scorecard-indicated outcome

1. Measurement or estimation of factors in the scorecard

In the "Discussion of the scorecard factors" section, we explain our analytical approach for scoring each scorecard factor or sub-factor, and we describe why they are meaningful as credit indicators. When a factor comprises sub-factors, we score at the sub-factor level. Some factors do not have sub-factors, in which case we score at the factor level.

The information used in assessing the sub-factors is generally found in or calculated from information in the company's financial statements or regulatory filings, derived from other observations or estimated by Moody's analysts. We may also incorporate non-public information.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends of a company's performance, as well as for peer comparisons. Financial metrics,[6] unless otherwise indicated, are typically calculated based on an annual or other 12-month period. However, the factors in the scorecard can be assessed using various time periods. For example, rating committees may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics incorporate our standard adjustments[7] to income statement, cash flow statement and balance sheet amounts for items such as underfunded pension obligations and operating leases. We may also make other analytical adjustments that are specific to a particular company.

2. Mapping scorecard factors to a numeric score

After estimating or calculating each factor or sub-factor, each outcome is mapped to a broad Moody's rating category (Aaa, Aa, A, Baa, Ba, B, Caa, or Ca, also called alpha categories), and to a numeric score.

Qualitative factors are scored based on the description by broad rating category in the scorecard. The numeric value of each alpha score is based on the scale below.

Exhibit 3

Numeric equivalents for qualitative factor and sub-factor scores

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

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Quantitative factors are scored on a linear continuum. For each metric, the scorecard shows the range by alpha category. We use the scale below and linear interpolation to convert the metric, based on its placement within the scorecard range, to a numeric score, which may be a fraction. As a purely theoretical example, if there were a ratio of revenue to interest for which the Baa range was 50x to 100x, then the numeric score for an issuer with revenue/interest of 99x, relatively strong within this range, would score closer to 7.5, and an issuer with revenue/interest of 51x, relatively weak within this range, would score closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the value of the metric that constitutes the lowest possible numeric score, and the value that constitutes the highest possible numeric score).

Exhibit 4

Ranges of numeric equivalents for quantitative factor and sub-factor scores

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

3. Determining the overall scorecard-indicated outcome

The numeric score for each sub-factor (or each factor, where the factor has no sub-factors), is multiplied by the weight for that sub-factor (or factor), with the results then summed to produce an aggregate numeric score. The aggregate numeric score is then mapped back to a scorecard-indicated outcome based on the ranges in the table below.

Exhibit 5

Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, an issuer with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the corporate family rating (CFR) for speculative-grade issuers and to the senior unsecured rating for investment-grade issuers. For issuers that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or Baseline Credit Assessment for comparison to

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the scorecard-indicated outcome. For an explanation of the Baseline Credit Assessment, please refer to *Rating Symbols and Definitions* and to our cross-sector methodology for government-related issuers.[8]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a corporate family rating (CFR) to speculative-grade issuers or a senior unsecured rating for investment-grade issuers. For issuers that benefit from rating uplift from government ownership, we may assign a Baseline Credit Assessment.[9]

Individual debt instrument ratings may be notched up or down from the CFR or the senior unsecured rating to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. The documents that provide broad guidance for such notching decisions are the rating methodology on loss given default for speculative-grade nonfinancial companies, the methodology for notching corporate instrument ratings based on differences in security and priority of claim, and the methodology for assigning short-term ratings.[10]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, please see *Rating Symbols and Definitions*.[11]

Limitations

In the preceding sections, we have discussed the scorecard factors and many of the other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall rating methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to actual ratings.

The scorecard in this rating methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings to these companies. Credit loss and recovery considerations, which are typically more important as an issuer gets closer to default, may not be fully captured in the scorecard. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for issuers at the upper and lower ends of the rating scale.

The weights for each factor and sub-factor in the scorecard represent an approximation of their importance for rating decisions across the sector, but the actual importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors that are outside of the scorecard, including those discussed above in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[12] Examples of such considerations include the following: how sovereign credit quality affects non-sovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking time periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology document does not include an exhaustive description of all factors that we may consider in assigning ratings in this sector. Companies in the sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for an issuer's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

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Moody's related publications

Credit ratings are primarily determined through the application of sector credit rating methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

For data summarizing the historical robustness and predictive power of credit ratings, please click here.

For further information, please refer to *Rating Symbols and Definitions*, which is available here.

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Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 For definitions of our most common metrics, please see our cross-sector methodology that describes our financial statement adjustments in the analysis of nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

4 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

5 For an explanation of the Baseline Credit Assessment, please refer to *Rating Symbols and Definitions* and to our cross-sector methodology for government-related issuers. A link to *Rating Symbols and Definitions* and a link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

6 For definitions of our most common metrics, please see our cross-sector methodology that describes our financial statement adjustments in the analysis of nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

7 For an explanation of our standard adjustments, please see our cross-sector methodology that describes our financial statement adjustments in the analysis of nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 A link to *Rating Symbols and Definitions* and a link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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10 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

11 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

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